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Day 02 Task: Packet Sniffing, Firewall Configuration, and Vulnerability Scanning

1.Objective

The objective of this task is to gain hands-on experience in core cybersecurity techniques including:

- Capturing and analyzing network traffic using Scapy.
- Implementing firewall rules with iptables.
- Performing vulnerability scanning using OpenVAS.

2. Tools and Technologies Used

Tool	Description
Scapy	A powerful Python-based interactive packet manipulation program.
iptables	A command-line firewall utility in Linux for packet filtering and NAT.
OpenVAS	Open-source vulnerability scanner used to detect system weaknesses.
Python 3	For scripting packet sniffing functionality using Scapy.
Linux (Kali/Ubuntu VM)	Test environment for configuring firewall and OpenVAS.
Matplotlib	Python library to generate bar chart of protocol distribution.

Tasks 1

1. Install Scapy:

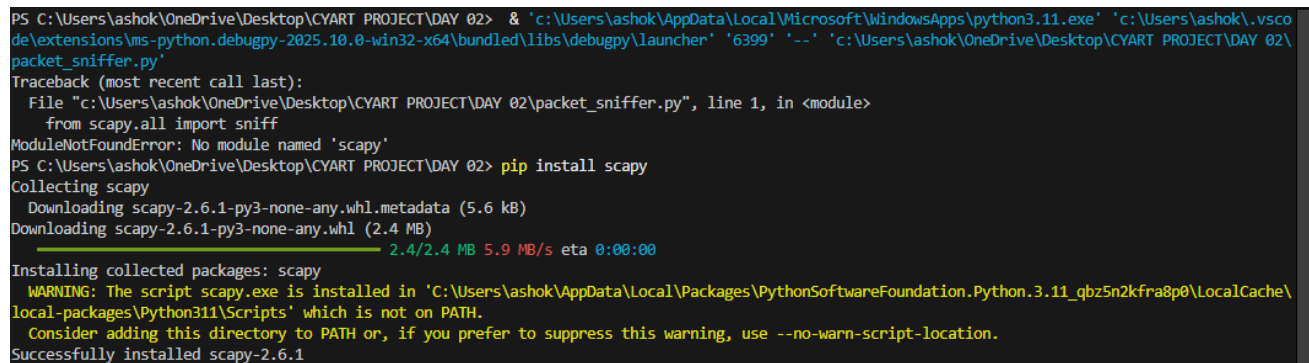
- Download and install Scapy from <https://scapy.net/>.

Step: 1

1. Download and install Scapy from <https://scapy.net/> or Clone Scapy from github (<https://github.com/secdev/scapy>) in Visual Studio Code and Terminal

Commands Used to Install Scapy:

1. pip install scapy in Visual Studio code



```
PS C:\Users\ashok\OneDrive\Desktop\CYART PROJECT\DAY 02> & 'c:\Users\ashok\AppData\Local\Microsoft\WindowsApps\python3.11.exe' 'c:\Users\ashok\.vscode\extensions\ms-python.debugpy-2025.10.0-win32-x64\bundled\libs\debugpy\launcher' '6399' '--' 'c:\Users\ashok\OneDrive\Desktop\CYART PROJECT\DAY 02\packet_sniffer.py'
Traceback (most recent call last):
  File "c:\Users\ashok\OneDrive\Desktop\CYART PROJECT\DAY 02\packet_sniffer.py", line 1, in <module>
    from scapy.all import sniff
ModuleNotFoundError: No module named 'scapy'
PS C:\Users\ashok\OneDrive\Desktop\CYART PROJECT\DAY 02> pip install scapy
Collecting scapy
  Downloading scapy-2.6.1-py3-none-any.whl.metadata (5.6 kB)
  Downloading scapy-2.6.1-py3-none-any.whl (2.4 MB)
    2.4/2.4 MB 5.9 MB/s eta 0:00:00
Installing collected packages: scapy
  WARNING: The script scapy.exe is installed in 'C:\Users\ashok\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5n2kfra8p0\LocalCache\local-packages\Python311\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed scapy-2.6.1
```

Fig1: Installing scapy Module Using Scapy Command in Visual Studio

Step: 2

1. Write a program code to analyse Packets and Capture the Packets that are present in the local Machine up to 100 Packets (Example Packets – TCP, UDP and OTHERS)

Program code :

```
from scapy.all import sniff

from scapy.layers.inet import IP, TCP, UDP, ICMP

from collections import Counter

packet_list = sniff(count=100)

protocols = []

for packet in packet_list:

    if packet.haslayer(TCP):

        protocols.append("TCP")

    elif packet.haslayer(UDP):

        protocols.append("UDP")

    elif packet.haslayer(ICMP):

        protocols.append("ICMP")

    else:

        protocols.append("Other")
```

```

protocol_count = Counter(protocols)

# Bar Chart

plt.bar(protocol_count.keys(), protocol_count.values(), color='skyblue')

plt.title("Protocol Distribution in Captured Packets")

plt.xlabel("Protocol")

plt.ylabel("Count")

plt.savefig("protocol_distribution.png")

plt.show()

```

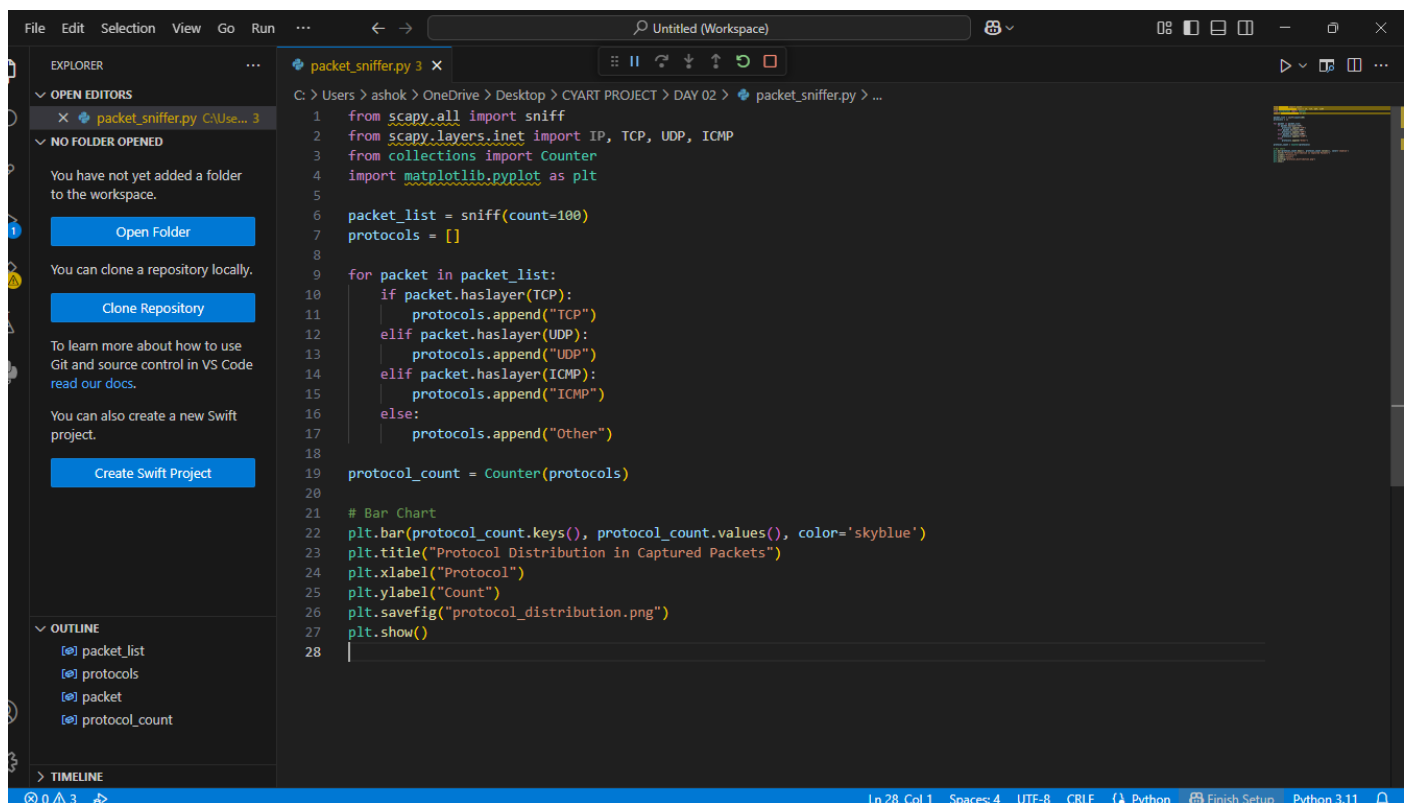


Fig 2 : Basic Python program to Analyse The Network Packets In Local Virtual Machine

Step 3 :

Lets Identify packets That are present in Local Virtual Machine

Note: To identify the Packets in local Virtual Machine .Before execute python program you need to install the python3.11 version in Your local Virtual Machine.

Trai 1 : Capturing Packets

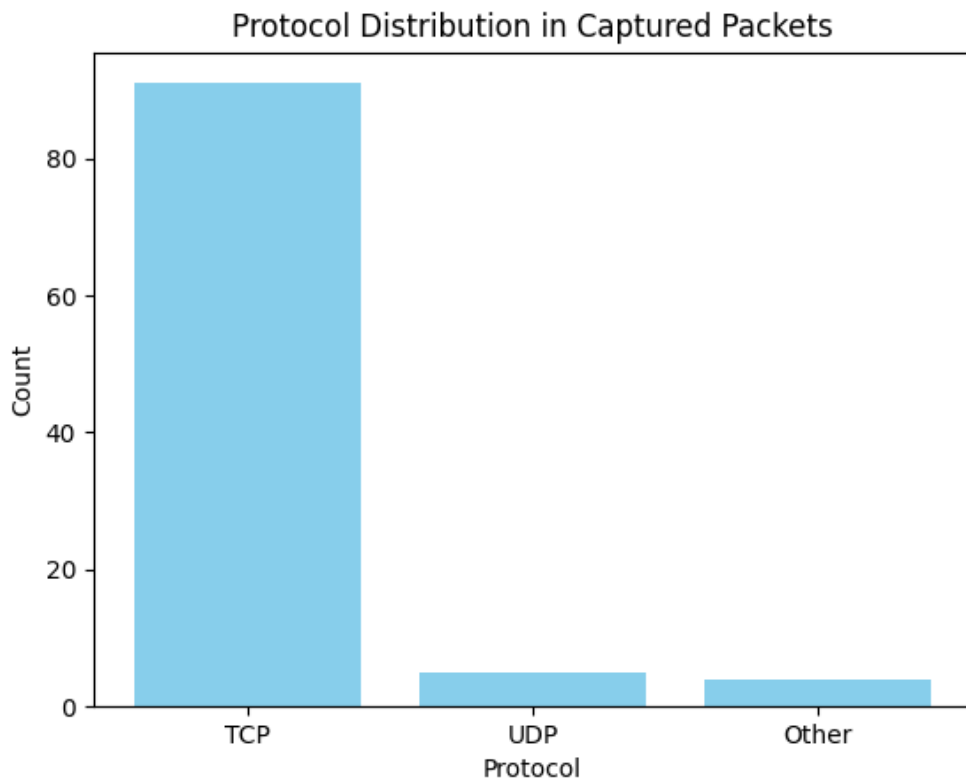


Fig 3: Different Protocols Distribution in the captured Packets. TCP packets are Transferring More than UDP and Other packets.

Trai 2 : Capturing Packets

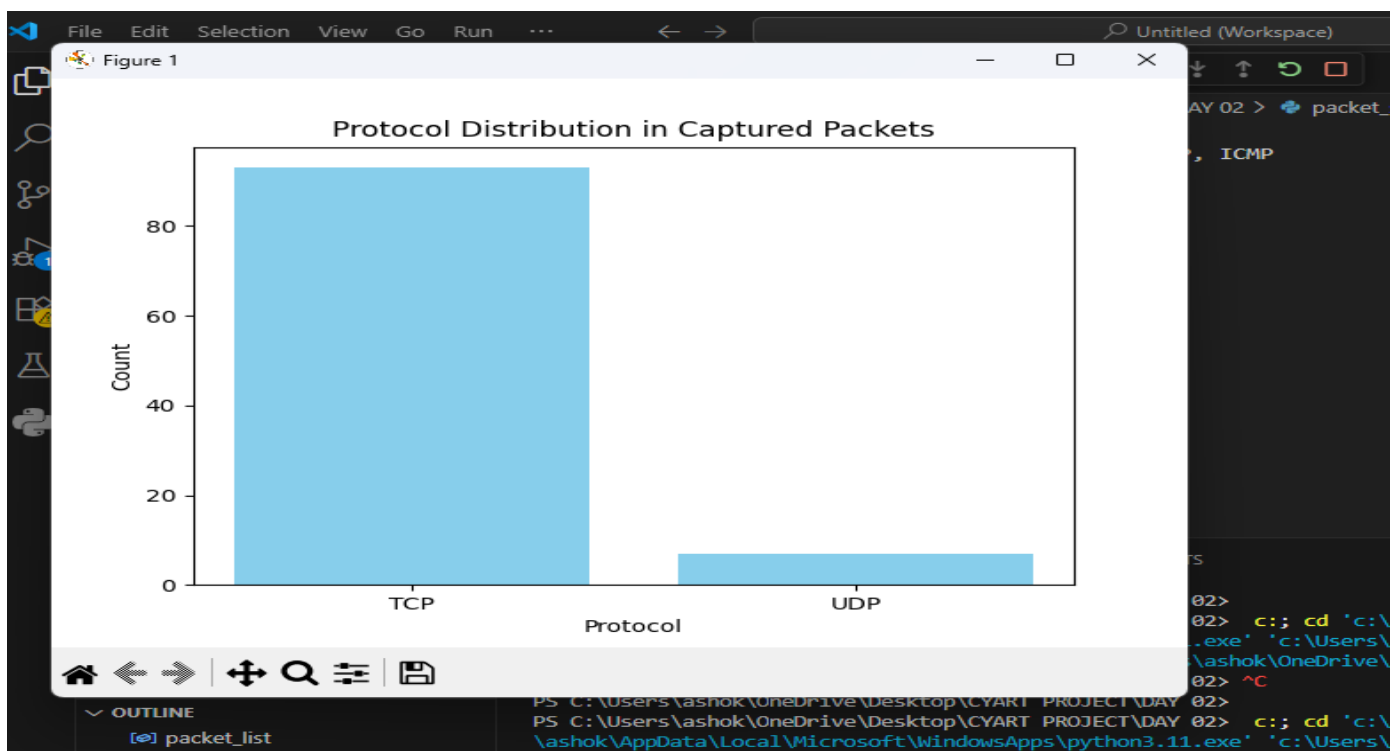


Fig 4: Different Protocols Distribution in the captured Packets. TCP packets are Transferring More than UDP

Trai 3 : Capturing Packets

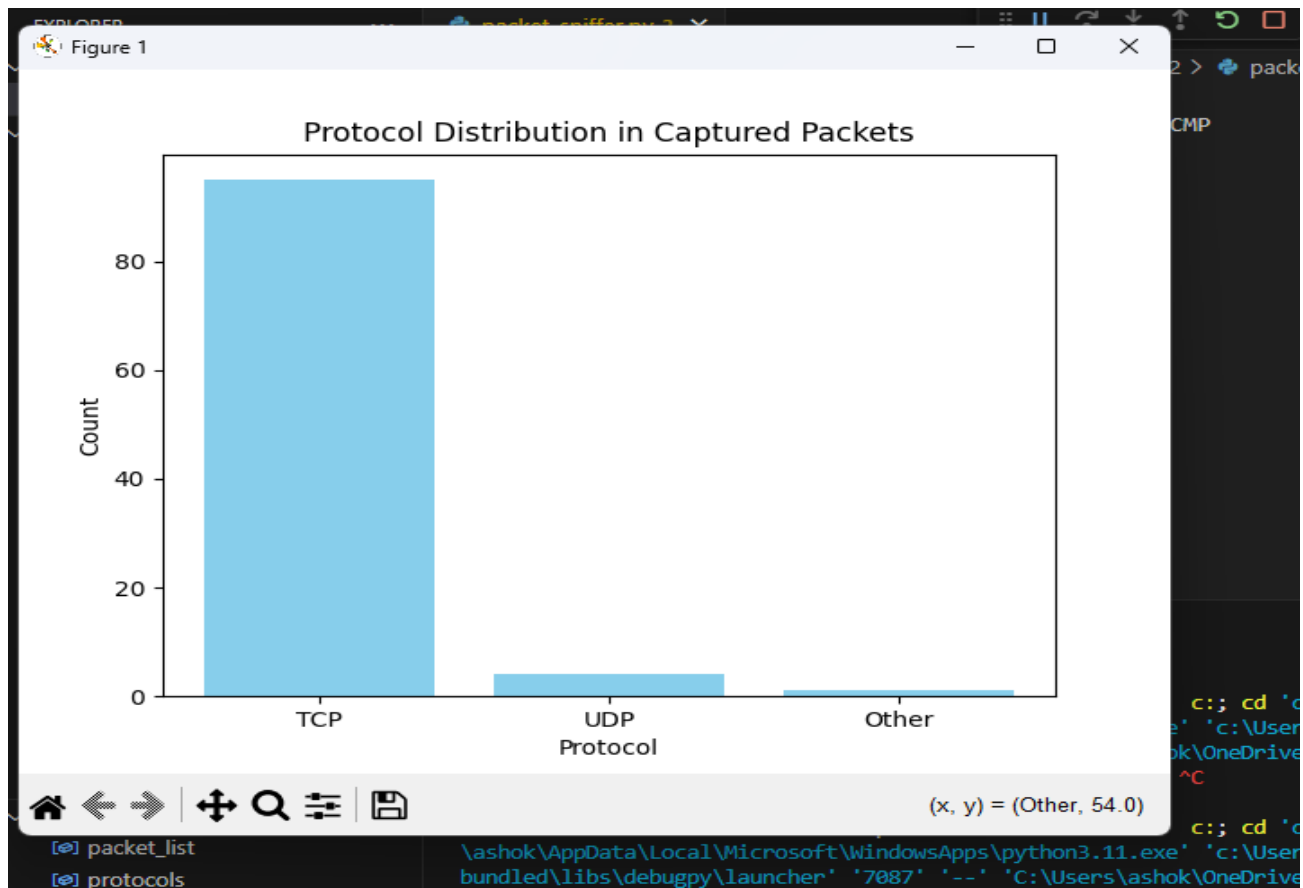


Fig 5: Different Protocols Distribution in the captured Packets. TCP packets are Transferring More than UDP and Other packets.

Results :

- Total Packets Captured: 100
- Protocols Identified: TCP, UDP, ICMP

Task 2 : Configure a Basic Firewall:

1. Use iptables to set up a basic firewall on a Linux machine (you can use a virtual machine if needed).
2. Allow SSH (port 22) and HTTP (port 80) traffic.
3. Block all other incoming traffic.
4. Test the firewall configuration by attempting to access blocked and allowed services.

2.1 What is Ip Tables?

iptables is a command-line firewall utility in Linux that uses rules to determine whether network traffic should be allowed, blocked, or forwarded. It operates on various tables and chains (INPUT, OUTPUT, FORWARD) and matches based on packet headers.

2.1.1 Create a Basic Firewall on Linux Virtual Machine:

Step: 1

- Open kali Linux virtual machine
- Open Terminal in kali linux
- Activate sudo or root in terminal

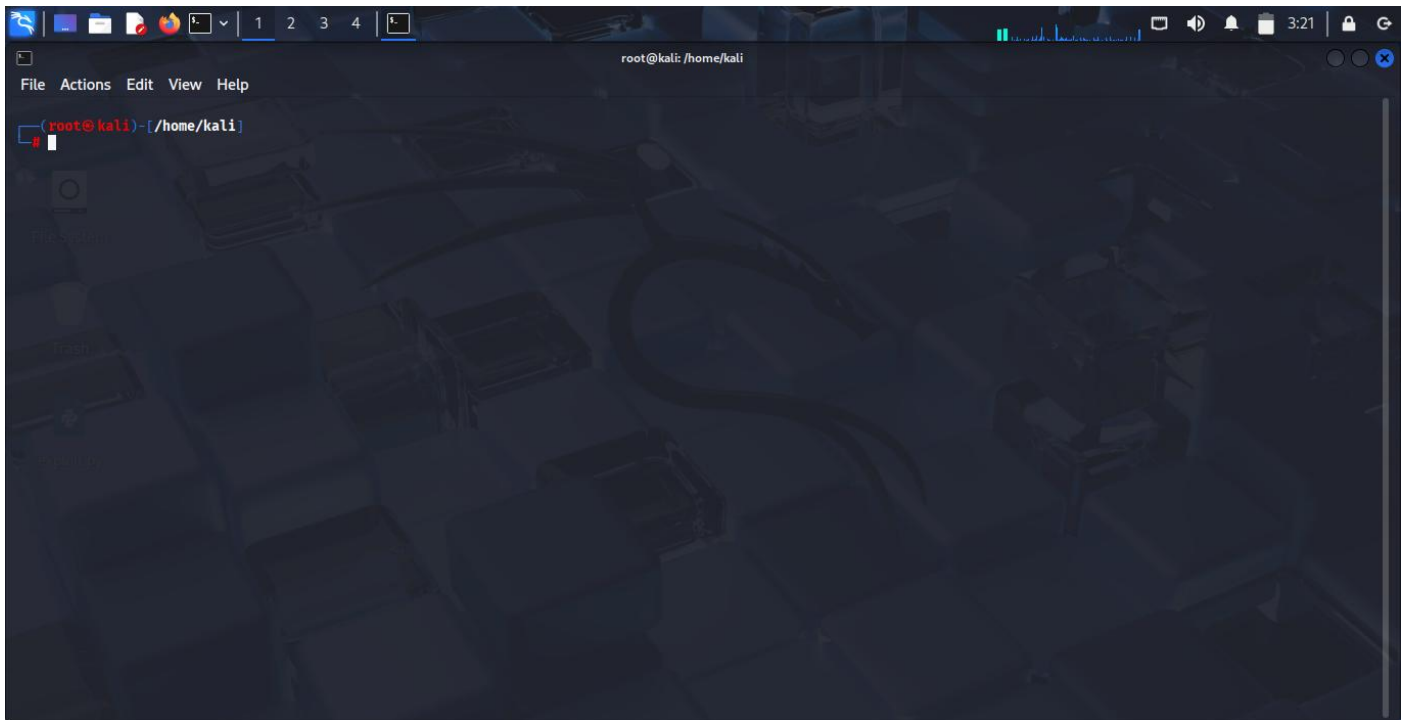


Fig 6 : Activating root in kali terminal

Step 2: Step up firewall Using iptables

Required Commands:

1.sudo iptables -F

(This command clears all existing rules in the default table.)

2.sudo iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT

(This rule allows responses to outgoing requests (like accessing a website from browser or ping replies).

3.sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

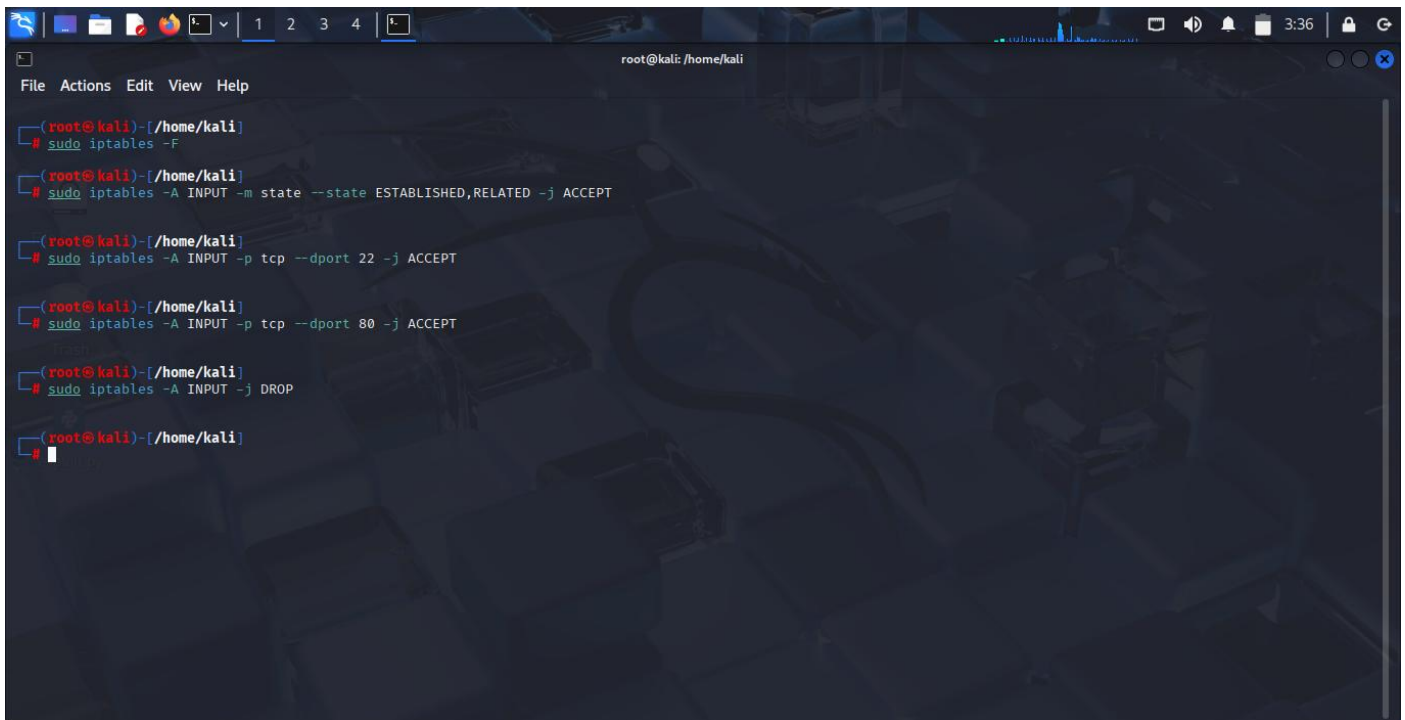
(Allows remote access via SSH (important if you're managing the system remotely).

4. sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT

(Allows incoming web traffic(HTTP), i.e., a basic website server.)

5.sudo iptables -A INPUT -j DROP

(This rule blocks everything else that wasn't allowed explicitly.)

A terminal window on a Kali Linux system showing a series of iptables commands being executed. The user is root at /home/kali. The commands are: 1. 'sudo iptables -F' to flush the table. 2. 'sudo iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT' to allow established connections. 3. 'sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT' to allow SSH. 4. 'sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT' to allow HTTP. 5. 'sudo iptables -A INPUT -j DROP' to drop all other incoming traffic. The terminal has a dark theme with a cityscape background.

```
root@kali: /home/kali
File Actions Edit View Help

(root@kali)-[/home/kali]
# sudo iptables -F

(root@kali)-[/home/kali]
# sudo iptables -A INPUT -m state --state ESTABLISHED,RELATED -j ACCEPT

(root@kali)-[/home/kali]
# sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

(root@kali)-[/home/kali]
# sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT

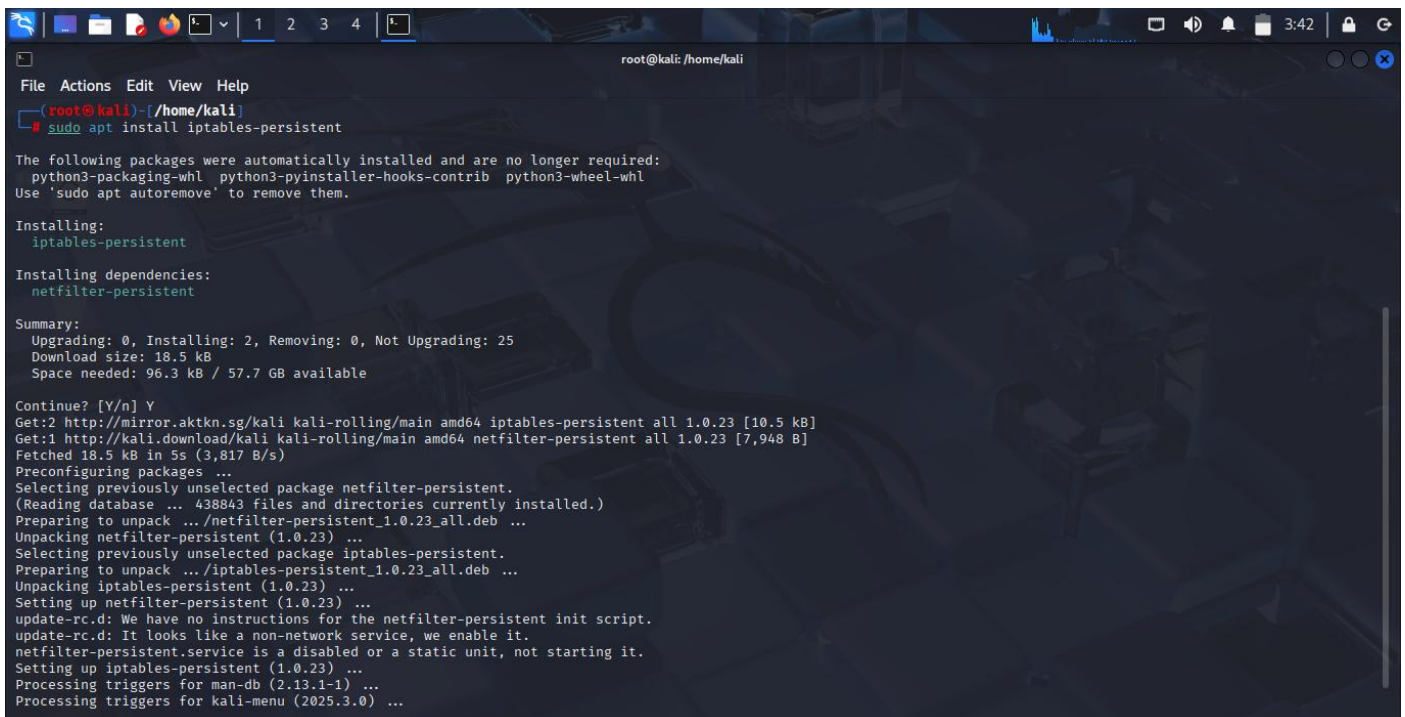
(root@kali)-[/home/kali]
# sudo iptables -A INPUT -j DROP

(root@kali)-[/home/kali]
#
```

Fig 7 Enter the all commands for step up the iptables in fire wall

Step 3: Install iptables-persistent

Command : Sudo apt install iptables-persistent

A terminal window showing the command 'sudo apt install iptables-persistent' being executed. The output shows that several python3 packages were automatically installed and are no longer required. The packages to be installed are 'iptables-persistent' and its dependency 'netfilter-persistent'. The summary shows that 2 packages will be installed, with a download size of 18.5 kB and a space needed of 96.3 kB. The user confirms the installation with 'Y'. The terminal shows the progress of downloading and installing the packages, including the setup of the netfilter-persistent init script and the iptables-persistent unit. The terminal has a dark theme with a cityscape background.

```
root@kali: /home/kali
File Actions Edit View Help

(root@kali)-[/home/kali]
# sudo apt install iptables-persistent

The following packages were automatically installed and are no longer required:
 python3-packaging-whl python3-pyinstaller-hooks-contrib python3-wheel-whl
Use 'sudo apt autoremove' to remove them.

Installing:
 iptables-persistent

Installing dependencies:
 netfilter-persistent

Summary:
 Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 25
 Download size: 18.5 kB
 Space needed: 96.3 kB / 57.7 GB available

Continue? [Y/n] Y
Get:2 http://mirror.aktkn.sg/kali kali-rolling/main amd64 iptables-persistent all 1.0.23 [10.5 kB]
Get:1 http://kali.download/kali kali-rolling/main amd64 netfilter-persistent all 1.0.23 [7,948 B]
Fetched 18.5 kB in 5s (3,817 B/s)
Preconfiguring packages ...
Selecting previously unselected package netfilter-persistent.
(Reading database ... 438843 files and directories currently installed.)
Preparing to unpack .../netfilter-persistent_1.0.23_all.deb ...
Unpacking netfilter-persistent (1.0.23) ...
Selecting previously unselected package iptables-persistent.
Preparing to unpack .../iptables-persistent_1.0.23_all.deb ...
Unpacking iptables-persistent (1.0.23) ...
Setting up netfilter-persistent (1.0.23) ...
update-rc.d: We have no instructions for the netfilter-persistent init script.
update-rc.d: It looks like a non-network service, we enable it.
netfilter-persistent.service is a disabled or a static unit, not starting it.
Setting up iptables-persistent (1.0.23) ...
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.3.0) ...
```

Fig 8 : successfully installed iptables-persistent

To save Iptables and Blocked all incoming Traffic

Command :

sudo iptables-save > /etc/iptables/rules.v4

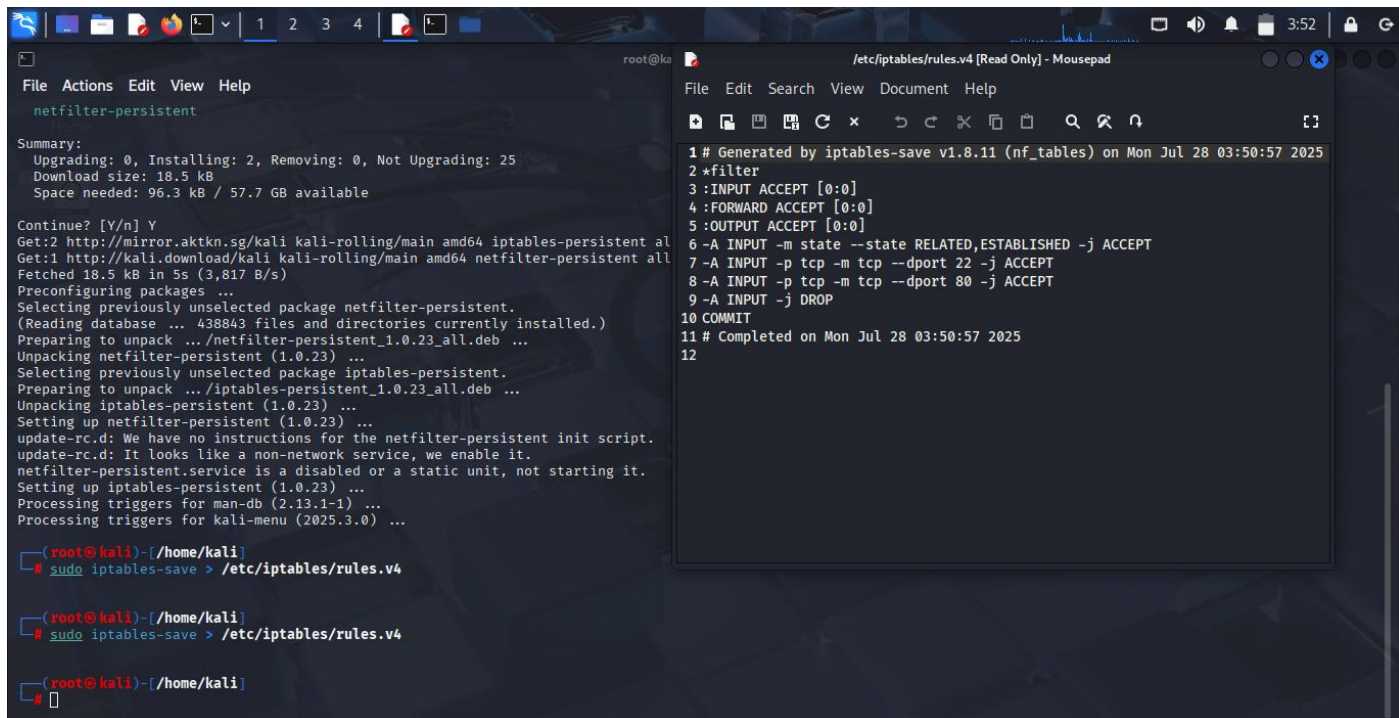


Fig 9 : Successfully iptables input, output, forward ACCEPT is Saved and Blocked incoming

Step 4:

Test the firewall configuration by attempting to access blocked and allowed services.

Protocol	Port	Test Command	Expected Result	Status
SSH	22	ssh user@192.168.1.100	Login prompt shown	Allowed
HTTP	80	curl http://192.168.1.100	HTML or response shown	Allowed
FTP	21	telnet 192.168.1.100 21	Connection refused or failed	Blocked
ICMP	N/A	ping 192.168.1.100	Timed out	Blocked

Fig 10 : filtered SSH and HTTP packet services are allowed

Task 3: Perform Vulnerability Scanning:

1. Install OpenVAS on a Linux machine (follow the setup guide for your distribution).
2. Configure OpenVAS and perform a scan on your local network or a test VM.
3. Analyze the scan results and document any vulnerabilities found, including their severity and potential impact.

3.1 What is OpenVAS?

OpenVAS (Open Vulnerability Assessment System) is a full-featured open-source vulnerability scanner maintained by Greenbone Networks. It is used to scan systems for known security flaws and misconfigurations.

Installation steps in kali linux:

Step 1 : update kali linux

```
sudo apt update && sudo apt upgrade -y
```

step 2 : Install OpenVAS (GVM)

```
sudo apt install openvas -y
```

Step 3 : Setup GVM

```
sudo gvm-setup
```

step 4 : Start GVM

```
sudo gvm-start
```



```
kali@kali: ~  
File Actions Edit View Help  
Scanner modified.  
+ Done  
* Please note the password for the admin user  
* User created with password '042149a5-4e3d-4e2c-869d-fb8d400b5355'.  
> You can now run gvm-check-setup to make sure everything is correctly configured  
--(kali@kali)-[~]  
$ sudo gvm-start  
sudo] password for kali:  
> Please wait for the GVM services to start.  
>  
> You might need to refresh your browser once it opens.  
>  
> Web UI (Greenbone Security Assistant): https://127.0.0.1:9392  
  
gsad.service - Greenbone Security Assistant daemon (gsad)  
Loaded: loaded (/usr/lib/systemd/system/gsad.service; disabled; preset: disabled)  
Active: active (running) since Mon 2025-07-28 05:48:02 EDT; 41ms ago  
Invocation: add890a564bf468eb6dedaca513d313a  
Docs: man:gsad(8)  
       https://www.greenbone.net  
Main PID: 23509 (gsad)  
Tasks: 3 (limit: 4546)  
Memory: 3.4M (peak: 3.4M)  
CPU: 33ms  
CGroup: /system.slice/gsad.service  
        └─23509 /usr/sbin/gsad --foreground --listen 127.0.0.1 --port 9392  
        └─23511 /usr/sbin/gsad --foreground --listen 127.0.0.1 --port 9392  
  
Jul 28 05:48:01 kali systemd[1]: Starting gsad.service - Greenbone Security Assistant daemon (gsad) ...  
Jul 28 05:48:02 kali gsad[23509]: gsad main:MESSAGE:2025-07-28 09h48.02 utc:23509: Starting GSAD version 24.2.3-git  
Jul 28 05:48:02 kali systemd[1]: Started gsad.service - Greenbone Security Assistant daemon (gsad).  
  
gvmd.service - Greenbone Vulnerability Manager daemon (gvmd)
```

Fig 10 : Once GVM start you will get the web Ui link click and start the Vulnerability Scan

Web UI (Greenbone Security Assistant): <https://127.0.0.1:9392>

User : Admin

password : at time of GVM start it will show the password. Please note it.

3.2 Performing a Vulnerability Scan

- Login to the Greenbone web portal.

- Go to Scans > Tasks > New Task.
- Enter a name (e.g., "Local Network Scan").
- In Target, specify:
An IP (e.g., 192.168.88.18) OR
A range (e.g., 192.168.88.18/24).
- Choose Full and Fast as the scan configuration.
- Click Create and then Start the scan.

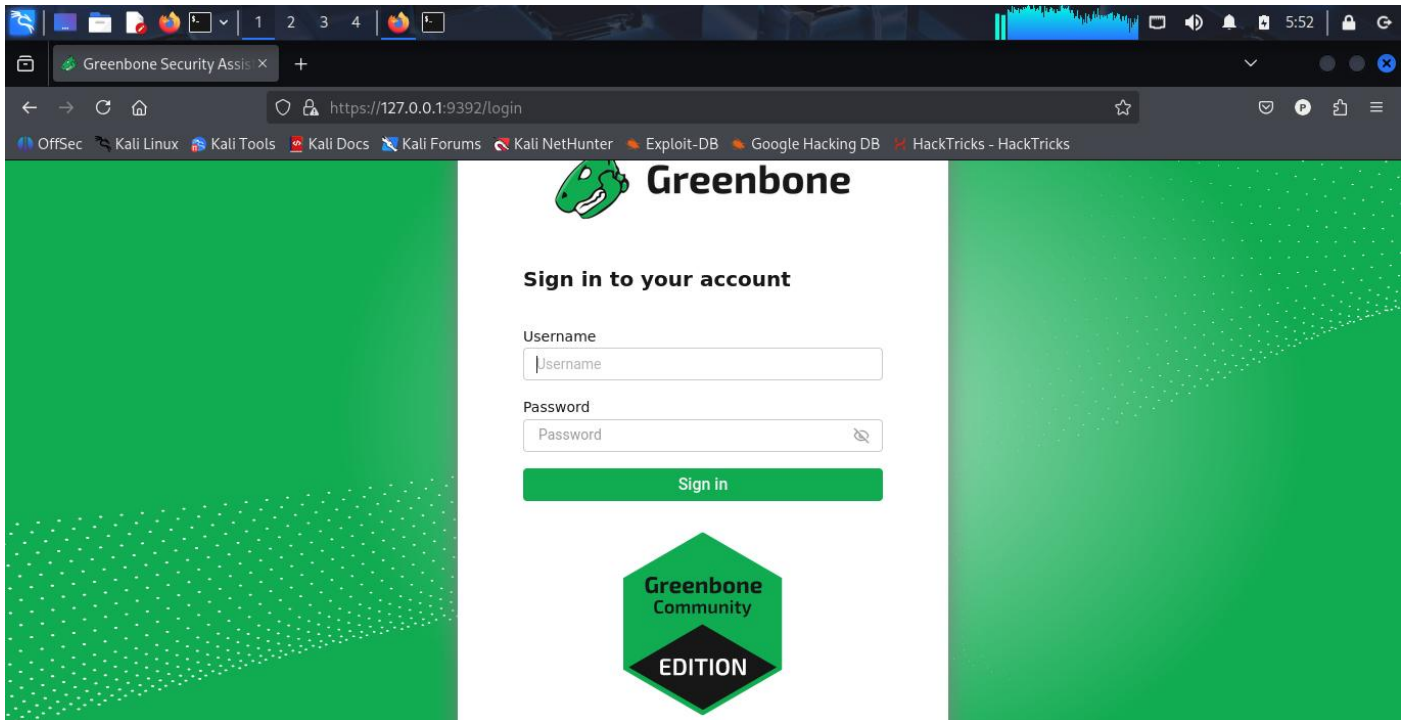


Fig 11 : once open the url you will see the login page and with username and password

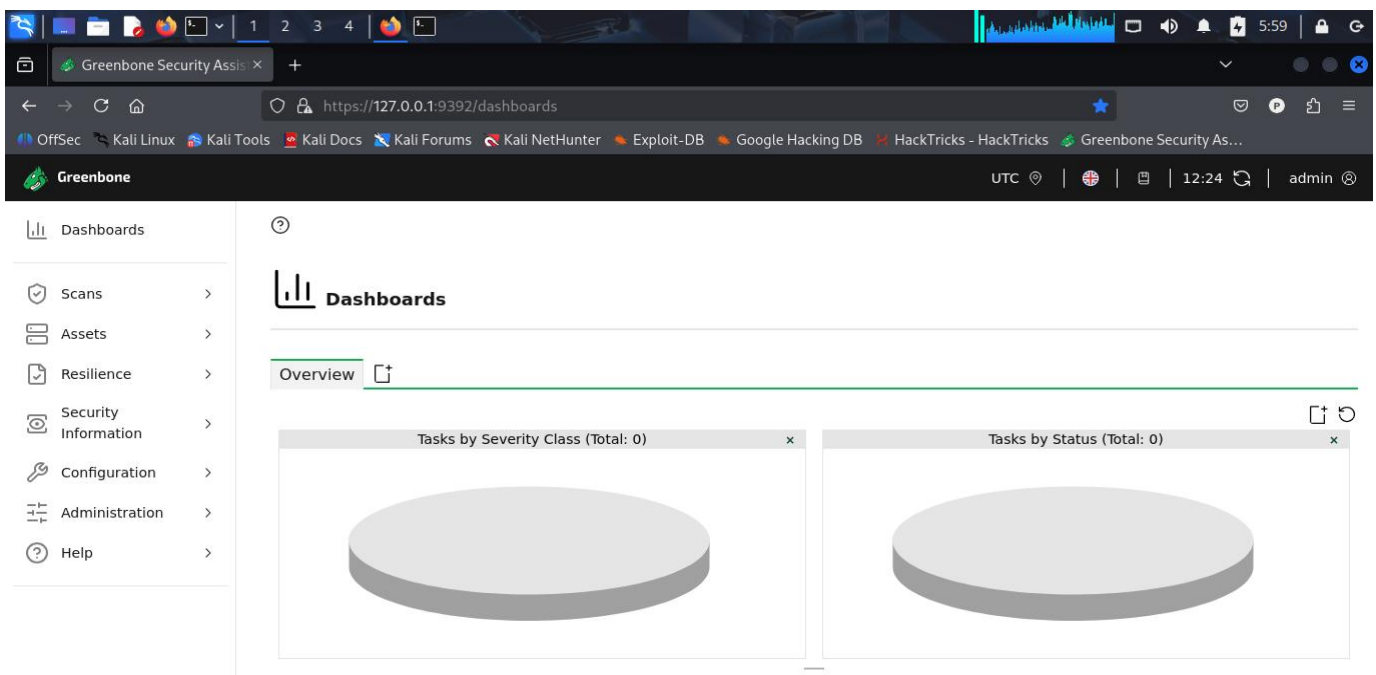


Fig 12 : Dashboard of Greenbone virtual Machine(GVM)

Task Wizard

Quick start: Immediately scan an IP address

IP address or hostname:

The default address is either your computer or your network gateway.
As a short-cut the following steps will be done for you:

1. Create a new Target
2. Create a new Task
3. Start this scan task right away

As soon as the scan progress is beyond 1%, you can already jump to the scan report by clicking on the bar in the "Status" column and review the results collected so far.

The Target and Task will be created using the defaults as configured in "My Settings".

Fig 13 : Quick scan an ip Address 192.168.88.18



Fig 14 : checking a task status

Advanced Task Wizard

Quick start: Create a new task

This wizard can help you by creating a new scan task and automatically starting it.

All you need to do is enter a name for the new task and the IP address or host name of the target, and select a scan configuration.

You can choose, whether you want to run the scan immediately, schedule the task for a later date and time, or just create the task so you can run it manually later.

In order to run an authenticated scan, you have to select SSH and/or SMB credentials, but you can also run an unauthenticated scan by not selecting any credentials. If you enter an email address in the "Email report to" field, a report of the scan will be sent to this address once it is finished.

For any other setting the defaults from "My Settings" will be applied.

Task Name

Scan Config

Target Host(s)

☐ Start immediately

☒ Create Schedule:

02/20/2023

at 5 h 0 m

Coordinated Universal Time/UTC

☐ Do not start automatically

SSH Credential on port

SMB Credential

ESXi Credential

Email report to

Fig 15 : advance Ip address Scan for advance task wizards

Analysing the Scan results:

- Go to Scans > Reports.
- Click the report name to open it.

Report Contains:

Field	Description
CVE ID	Common Vulnerability Enumeration reference
Severity	Ranked 0 (None) to 10 (Critical)
Host Affected	IP of the vulnerable system
Summary	Short description of the issue
Solution	Suggested remediation

Vulnerability found in 192.168.88.18

CVE ID	Severity	Description	Impact	Solution
CVE-2021-41773	High (9.8)	Apache Path Traversal	Remote Code Execution	Upgrade Apache
CVE-2017-0143	Critical (10.0)	SMBv1 RCE (EternalBlue)	WannaCry risk	Disable SMBv1, Patch system
CVE-2020-0601	Medium (5.9)	Weak crypto in Windows	MITM	Apply Microsoft patch

References:

Scapy Documentation: <https://scapy.net/>

Iptables man page : <https://man7.org/linux/man-pages/man8/iptables.8.html>

Digital Ocean tables : <https://www.digitalocean.com/community/tutorials/iptables-essentials-common-firewall-rules-and-commands>

Open VAS Documentation : <https://greenbone.github.io/>